

DR. BHAVANA MERUGU

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Summary

Experienced Data Analyst with 4+ years in the healthcare sector, skilled in deriving actionable insights from complex datasets to enhance patient care and operational efficiency. Proficient in SQL, Python, and R for data analysis, and adept at utilizing Tableau and Power BI for visualization and reporting. Experienced with Electronic Health Records (EHR) and Health Information Systems (HIS), with a strong focus on predictive analytics, data warehousing, and ETL processes. Demonstrates expertise in data quality and governance, and excels in translating data into strategic recommendations.

Technical Skills

Languages: Python, SQL, R, DAX, HTML, CSS

Frameworks & Tools: NumPy, Pandas, Matplotlib, GitHub, Docker, Statistics, Agile, SAS, SSIS, PySpark, ggplot2, Snowflake, Seaborn, Sci-kit Learn, Ad hoc, KPI Analysis, Automation Power Query

Databases: MySQL, Redshift, Snowflake, Big Query, PostgreSQL, MongoDB, SparkSQL

Data Analysis / Tools: Power BI, Tableau, Data Bricks, MS Excel, Google Analytics, Data (Modeling, Mapping, Mining, Extraction, and Transformation), Jira, Confluence

Healthcare Skills: EMR/HER, Medical Coding Systems (ICD-10-CM, CPT), HIPAA Compliance, PHI, PII, FDA, OSHA, HITECH, Claims Submission Processes, Medicare and Medicaid Billing, Patient Advocacy, Contact center experience, HL7

Professional Experience

Data Analyst, Mayo Clinic

07/2024 – Present | USA

- Leveraged Python (NumPy, Pandas) and SQL to optimize data pipelines, reducing query execution time by 35%, ensuring faster access to Electronic Health Records (EHR) and Clinical Decision Support Systems (CDSS), directly improving clinical workflows and patient outcomes.
- Improved the performance of healthcare databases hosted on Amazon RDS by restructuring SQL schemas, resulting in 40% faster queries and better supporting hospital administration, billing, and compliance operations.
- Automated Python-driven data validation checks for 250,000+ daily patient records, achieving full HIPAA compliance, enhancing data accuracy, and reducing inconsistencies by 30% through robust validation frameworks.
- Implemented real-time analytics solutions using Python and SQL to deliver dynamic dashboards and reports, enhancing clinical decision-making and hospital resource planning, and improving operational efficiency by 20%.
- Automated healthcare operations monitoring, reducing system downtime by 25% while improving patient record access, inventory tracking, and compliance reporting for enhanced system availability and reliability.
- Developed interactive Power BI dashboards for hospital leadership, providing real-time insights into patient health and hospital performance, improving decision-making, and increasing administrative efficiency by 25%.
- Collaborated with healthcare stakeholders to optimize workflows, medical records management, and revenue cycles while ensuring compliance with CMS regulations and ICD-10 standards, resulting in improved clinical operations and financial performance.

Data Analyst, Apollo Hospitals

04/2020 – 05/2023 | India

- Spearheaded a comprehensive healthcare data optimization project, managing healthcare administration for a 500+ patient facility by utilizing Electronic Health Records (EHR) and Practice Management Systems (PMS) to streamline workflows and improve operational efficiency.
- Implemented a chronic disease management program by analyzing large datasets from EHRs, insurance claims, and patient surveys, leading to a 12% increase in patient satisfaction and a 5% reduction in overall healthcare costs.
- Utilized SAS and SQL to enhance healthcare analytics, improving reporting processes and delivering actionable insights, which resulted in a 15% increase in patient satisfaction and a 10% reduction in response times.
- Conducted rigorous data quality assessments using Informatica and Talend, ensuring a 20% improvement in data integrity, which enabled more accurate clinical and financial analysis.
- Established KPIs for patient outcomes and healthcare efficiency, resulting in a 30% increase in adherence to clinical guidelines and promoting a culture of data-driven decision-making.
- Developed dynamic Tableau dashboards, reducing report generation time by 30%, enabling real-time insights for hospital administrators and stakeholders to optimize resource allocation and improve patient care quality.

Education

Masters of Science, Franklin University

Healthcare Informatics

Bachelors in Dental Surgery, Mamata Dental College